

Dry Run

① CallStackTrace.java

	Stack (top → bottom)
Call factorial(4)	4
Call factorial(3)	3, 4
Call factorial(2)	2, 3, 4
Call factorial(1), base case returns 1	1, 2, 3, 4
factorial(2) = $2 * 1 = 2$	2, 3, 4
factorial(3) = $3 * 2 = 6$	3, 4
factorial(4) = $4 * 6 = 24$	4

Output → factorial(4) frame pushed, stack depth = 1
 → factorial(3) frame pushed, stack depth = 2
 → factorial(2) frame pushed, stack depth = 3
 → factorial(1) frame pushed, stack depth = 4

← factorial(1) returns 1 frame popped
 ← fact(2) returns 2 frame popped
 ← fact(3) returns 6 frame popped
 ← fact(4) returns 24 frame popped
 main receives 24

② Direct Indirect

isEven(10)
 ↓
 isOdd(9)
 ↓

isEven(8)

isEven(0)

true

~~3~~ n goes down by each time and 2 methods keep swapping

The base case is $n=0$ isEven(0) is true & isOdd(0) is false

Head - head(3) → head(2) → head(1) → head(0) stops
print while coming back: 1, 2, 3

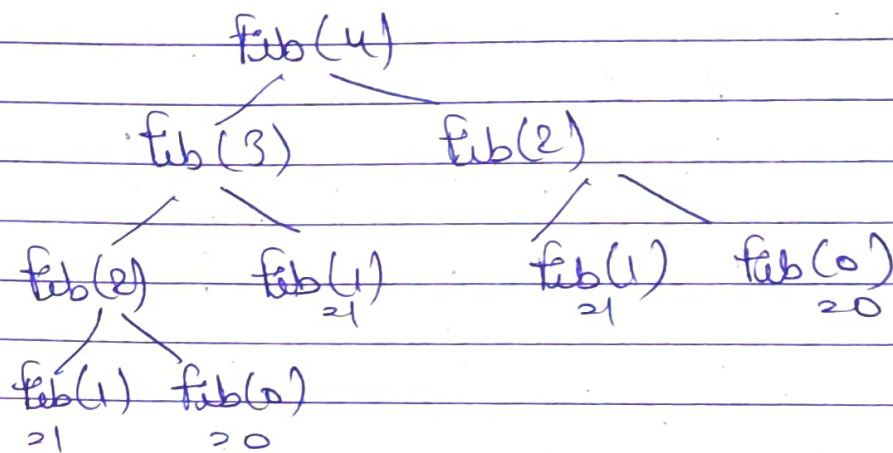
Tail - tail(3) print(3) → tail(2) print(2) → tail(1)
print(1) → tail(0) stops
o/p 3 2 1

Tail Recursive factorial : factTail(4, 1)

call	an
factTail(4, 1)	1
factTail(3, 4)	4
factTail(2, 12)	12
factTail(1, 24)	24, so return 24

Tree Recursion = fib(4)

$$\begin{aligned}
 \text{fib}(4) &= \text{fib}(3) + \text{fib}(2) \\
 &= (\text{fib}(2) + \text{fib}(1)) + (\text{fib}(1) + \text{fib}(0)) \\
 &= ((1+0) + 1) + (1+0) \\
 &= 3
 \end{aligned}$$



9 calls for $n=4$ 21 calls for $n=20$

③ Fibonacci Compare

iterative (5), start $a=0$, $b=1$

i	t	a	b
0	1	1	1
1	2	1	2
2	3	2	3
3	5	3	5
4	8	5	8

return a=5

output (n=5, 7, 9)

n	fib(n)	naive calls	memo calls	iter
5	5	15	9	5
7	13	41	13	13
9	34	109	17	34

④ Java Recursion Gotcha

Self referencing list, the list contains itself :-
 (x, (this Collection))

calling hashCode() repeatedly calls itself
 result - StackOverflowError

Employee

Initially:

Asha → Ravi → null

after b. manages = a

Asha → Ravi → Asha → Ravi → ...

This never ends

Result: StackOverflowError

5. Recurrence Counter

divide And Conquer (4)

work $\rightarrow 4$

left (2) $\rightarrow +2$

right (2) $\rightarrow +2$

Total $= 4 + 2 + 2 = 8$

so $n \log_2 n = 4 \times 2 = 8$

values

n	break	halving	divide & Conquer
1024	1024	10	10240
2048	2048	11	22528
4096	4096	12	49152
8192	8192	13	106496

$n=10 \rightarrow 1023$

$n=15 \rightarrow 32767$

$n=20 \rightarrow 1048575$

formula $2^n - 1$

6. Recursion Basics

Countdown

5 → 4 → 3 → 2 → 1

Output

5 4 3 2 1 Go!

Factorial $10 \times 9 \times 8 \times \dots \times 1 = 3628800$ Sum $3 + 1 + 4 + 1 + 5 = 14$ Sum of digits $4721 \rightarrow 1 + 2 + 7 + 4 \rightarrow 14$ Low power $2^{30} = 1073741824$ Fast power $= 62 \rightarrow 31 \rightarrow 15 \rightarrow 7 \rightarrow 3 \rightarrow 1 \rightarrow 0$
 $2^{62} = 4611686018427387904$ Reverse - REURSTON → NOTSRUCBPalindrome - MALAYALAM

M = M

A = A

L = L

A = A

true

GCD $(252, 105)$
→ $(105, 42)$
→ $(42, 21)$
→ $(21, 0)$ GCD = 21

Binary Search

Search = 23

mid = 4 → 16 → go right

mid = 7 → 56 → go left

mid = 5 → 23 → Found

Index = 5

decimal to Binary 37 → 18 → 9 → 4 → 2 → 1 → 0

Remainders Reversed: 100101

⑦ Recursive Linked List

list: 10 → 20 → 30 → 40 → null

Backward:

40 30 20 10

length = 4

contains 30 = true

Reverse :- 10 original 10 20 30 40

recursion reaches 40

then links change while returning

40 → 30

40 → 30 → 20

40 → 30 → 20 → 10

Final: 40 30 20 10 null

⚠ very large list with may cause StackOverflow Error.

⑧ Stack Depth

1. no BaseLine

call $\rightarrow$ call $\rightarrow$ call $\rightarrow$...

and result will be StackOverflowError

2. big frame - Each function call uses a larger stack frame

so overflow happens sooner

3. Wrong Base (-1)

-1 $\rightarrow$ -2 $\rightarrow$ -3 $\rightarrow$ -4 $\rightarrow$...

it never reaches 0

Result - StackOverflowError

4. 64MB thread - more stack memory $\rightarrow$ more calls possible before overflow

⑨ Subsets/Permutations

Subsets of ABC, for every character, there are 2 choices
Include / exclude

so $2^3 = 8$

Subsets :-

- ABC
- AB
- AC
- A
- BC
- B
- C
- {} (empty set)

Permutations :- ABC, ACB, BAC, BCA, CAB, CBA
 no. $3! = 6$, for 8 letters $8! = 40320$

⑩ Tower of Hanoi

for ~~n=3~~ $n=3$;

1. $A \rightarrow C$
2. $A \rightarrow B$
3. $C \rightarrow B$
4. $A \rightarrow C$
5. $B \rightarrow A$
6. $B \rightarrow C$
7. $A \rightarrow C$

total - 7 moves formula $2^n - 1$

n	moves
5	31
10	1023
15	32767
20	1048575
25	33554431

$n=64$ $2^{64} - 1$

⑪ temp.java

table(2,3) - print before recursive call: table(2,3) → 6
 tables(2,4) → 8, tables(2,5) → 10, tables(2,10) → 20
 tables(2,11) → return
 o/p - 6, 8, 10, 12, 14, 16, 18, 20

2) printSeries(1) - call let print after recursive call 1 → 2 → 3 → ...
 10 → 11
 at 11, return starts
 while returning - 10 9 8 7 6 5 4 3 2 1
 print after call → reverse order

3) diffOfTwo(0)
 0 → 2 → 4 → 6 → ... → 20 → 22
 at 22, return
 o/p 0 2 4 6 8 10 12 14 16 18 20

Print → call = same order
 Call → Print = reverse order